

Technical Note

PlatoSim Reference Frames

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Document History

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1	2	2015-09-01	Sign corrections, so that all rotations are now clockwise.
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4	1	2016-05-24	Added a small section the camera drift rotation angles.
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9	1	2022-10-04	Changed the spacecraft RF to comply with PLATO-DLR-MIS-H0-0006-v3.
9	2	2022-10-18	Changed SC and TL nomenclature to PLM and CAM
9	3	2023-02-07	Changed rot matrix of PLM ref frame

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1 Abstract

2 Documents and Acronyms

2.1 Applicable Documents

AD1: PLATO-OHB-PL-LI-009, *PLATO Instrument Goordinate Systems*

AD2: PLATO-DLR-PL-TN-016, *PLATO Geometric Camera Model*

AD3: PLATO-DLR-MIS-HO-0006, *Clarifications on quaternions*

2.2 Reference Documents

RD1: P. Marcos-Arenal et al., 2014, A&A 566, A92, *The PLATO Simulator: Modelling of High-Cadence Space-Based Imaging*

2.3 Acronyms

CAM	Camera
CCD	Charge-Coupled Device
EQ	Equatorial reference frame
FP	Focal Plane
NP	Celestial Equatorial North Pole
PLM	Payload Module
PSF	Point Spread Function
SVM	Service Module

3 The Payload Module reference frame

The base reference system of PlatoSim is the standard right-handed celestial equatorial coordinate system $(X_{\text{EQ}}, Y_{\text{EQ}}, Z_{\text{EQ}})$ where X_{EQ} points towards the vernal equinox, and Z_{EQ} to the equatorial North Pole, as shown in Fig. 1.

We define a new right-handed cartesian coordinate system $(X_{\text{PLM}}, Y_{\text{PLM}}, Z_{\text{PLM}})$ which we will call the Payload Module reference frame¹, by performing the following rotations starting from the equatorial reference frame $(X_{\text{EQ}}, Y_{\text{EQ}}, Z_{\text{EQ}})$:

1. we rotate over an angle α_{PLM} around the Z_{EQ} axis, to obtain the $(X_{\text{A}}, Y_{\text{A}}, Z_{\text{A}})$ reference frame where $Z_{\text{A}} = Z_{\text{EQ}}$. This rotation is illustrated in Figure 1, and the corresponding passive rotation matrix is (cf. Eq. (37) and taking the transpose in Eq. (42) already into account):

$$\mathbf{R}_{\text{EQ}}^{\text{A}} = \begin{pmatrix} \cos \alpha_{\text{PLM}} & \sin \alpha_{\text{PLM}} & 0 \\ -\sin \alpha_{\text{PLM}} & \cos \alpha_{\text{PLM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (1)$$

while the unit quaternion is according to Eq. (56):

$$\mathbf{q}_{\text{EQ}}^{\text{A}} = \cos \frac{\alpha_{\text{PLM}}}{2} + \sin \frac{\alpha_{\text{PLM}}}{2} \mathbf{k} \quad (2)$$

2. we rotate over an angle $\pi/2 - \delta_{\text{PLM}}$ around the Y_{A} axis, to obtain the $(X_{\text{B}}, Y_{\text{B}}, Z_{\text{B}})$ reference frame where $Y_{\text{B}} = Y_{\text{A}}$. This rotation is illustrated in Figure 1 and the corresponding passive rotation matrix is (cf. Eqs. (36) and (42)):

$$\mathbf{R}_{\text{A}}^{\text{B}} = \begin{pmatrix} \sin \delta_{\text{PLM}} & 0 & -\cos \delta_{\text{PLM}} \\ 0 & 1 & 0 \\ \cos \delta_{\text{PLM}} & 0 & \sin \delta_{\text{PLM}} \end{pmatrix} \quad (3)$$

while the unit quaternion is

$$\mathbf{q}_{\text{A}}^{\text{B}} = \cos \left(\frac{\pi}{4} - \frac{\delta_{\text{PLM}}}{2} \right) + \sin \left(\frac{\pi}{4} - \frac{\delta_{\text{PLM}}}{2} \right) \mathbf{j} \quad (4)$$

The reason for $\pi/2 - \delta_{\text{PLM}}$ rather than δ_{PLM} is that δ_{PLM} is measured from the equator rather than the North Pole, while the rotation angle should increase clockwise when looking in the positive direction of the rotation axis.

3. we rotate over a roll angle κ around the Z_{B} axis, to obtain the $(X_{\text{PLM}}, Y_{\text{PLM}}, Z_{\text{PLM}})$ reference frame, where $Z_{\text{PLM}} = Z_{\text{B}}$. This rotation is illustrated in Figure 2, and the corresponding passive rotation matrix is (cf. Eqs. (37) and (42)):

$$\mathbf{R}_{\text{B}}^{\text{PLM}} = \begin{pmatrix} \cos \kappa & \sin \kappa & 0 \\ -\sin \kappa & \cos \kappa & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (5)$$

while the unit quaternion is

$$\mathbf{q}_{\text{C}}^{\text{PLM}} = \cos \frac{\kappa}{2} + \sin \frac{\kappa}{2} \mathbf{k} \quad (6)$$

The right-handed cartesian coordinate system $(X_{\text{PLM}}, Y_{\text{PLM}}, Z_{\text{PLM}})$ is associated with the spacecraft, in the same way as defined in [AD1]. Its origin is the geometric center of the interface between the bottom of the optical bench and the service module. The positive roll axis Z_{PLM} points towards the operator-given mean payload line-of-sight, given by the equatorial sky coordinates $(\alpha_{\text{PLM}}, \delta_{\text{PLM}})$. The X_{PLM} axis points in the direction of the highest point of the sunshield, as shown in the right-hand side of Fig. 2.

¹For historical reason, inside PlatoSim's code the payload module reference frame is called the spacecraft reference frame.

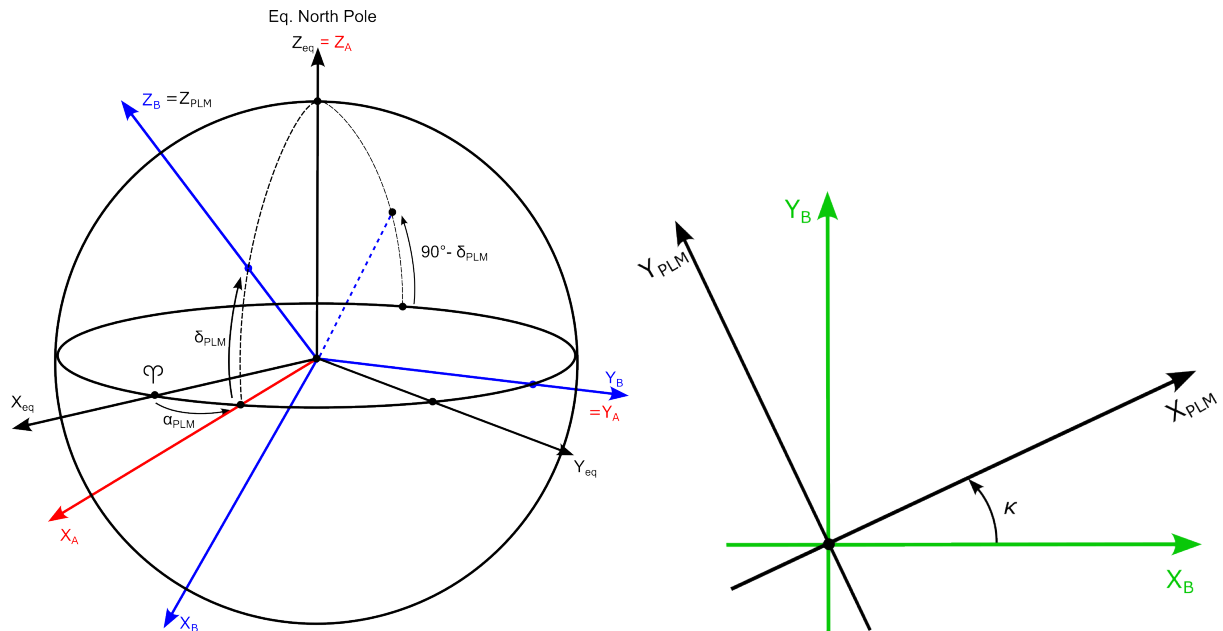


Figure 1: Illustration of the rotation between the reference frames $EQ \rightarrow A \rightarrow B \rightarrow PLM$. Left: The base reference frame is the celestial equatorial coordinate system (X_{EQ}, Y_{EQ}, Z_{EQ}) . The positive Z_B axis points towards the operator-given payload module pointing coordinates $(\alpha_{PLM}, \delta_{PLM})$. Right: The transformation between the (X_B, Y_B, Z_B) and the $(X_{PLM}, Y_{PLM}, Z_{PLM})$. The positive $Z_B = Z_{PLM}$ axis points towards the reader.

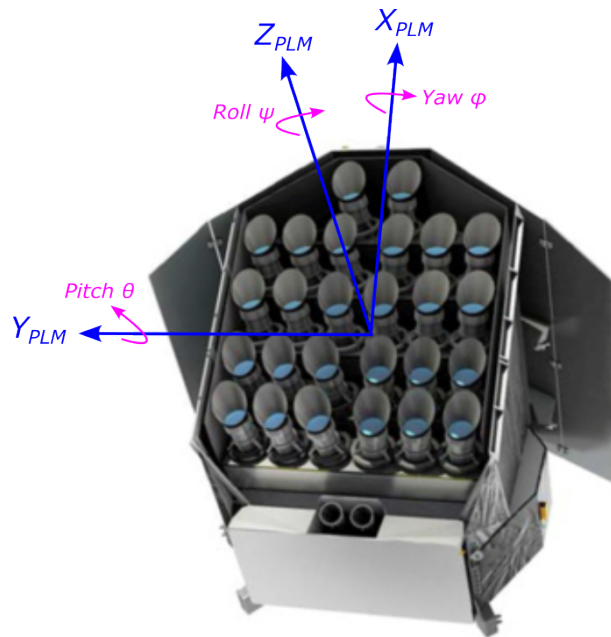


Figure 2: association of the right-handed $(X_{PLM}, Y_{PLM}, Z_{PLM})$ coordinate system to the payload module. The positive Z_{PLM} axis points towards the operator-given payload module pointing coordinates $(\alpha_{PLM}, \delta_{PLM})$. The X_{PLM} axis points in the direction of the highest point of the sunshield.

Given a target with celestial equatorial sky coordinates (α, δ) , its cartesian coordinates $(v_x, v_y, v_z)_{\text{PLM}}$ in the spacecraft reference frame are computed using the following transformation using rotation matrices:

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}_{\text{PLM}} = \mathbf{R}_{\text{EQ}}^{\text{PLM}}(\alpha_{\text{PLM}}, \delta_{\text{PLM}}, \kappa) \begin{pmatrix} \cos \delta \cos \alpha \\ \cos \delta \sin \alpha \\ \sin \delta \end{pmatrix}_{\text{EQ}} \quad (7)$$

where the rotation matrix $\mathbf{R}_{\text{EQ}}^{\text{PLM}}(\alpha_{\text{PLM}}, \delta_{\text{PLM}}, \kappa)$ is given by

$$\mathbf{R}_{\text{EQ}}^{\text{PLM}}(\alpha_{\text{PLM}}, \delta_{\text{PLM}}, \kappa) = \mathbf{R}_{\text{B}}^{\text{PLM}} \cdot \mathbf{R}_{\text{A}}^{\text{B}} \cdot \mathbf{R}_{\text{EQ}}^{\text{A}} \quad (8)$$

where the rotation matrices $\mathbf{R}_{\text{EQ}}^{\text{A}}$, $\mathbf{R}_{\text{A}}^{\text{B}}$, and $\mathbf{R}_{\text{B}}^{\text{PLM}}$ are given by Eqs. (1), (3), and (5).

For the quaternions we have, according to Eq. (62):

$$\begin{pmatrix} 0 \\ v_x \\ v_y \\ v_z \end{pmatrix}_{\text{PLM}} = \mathbf{q}_{\text{EQ}}^{\text{PLM}} * \begin{pmatrix} 0 \\ \cos \delta \cos \alpha \\ \cos \delta \sin \alpha \\ \sin \delta \end{pmatrix}_{\text{EQ}} \mathbf{q}_{\text{EQ}}^{\text{PLM}} \quad (9)$$

where the quaternion $\mathbf{q}_{\text{EQ}}^{\text{PLM}}$ is given by

$$\mathbf{q}_{\text{EQ}}^{\text{PLM}} = \mathbf{q}_{\text{EQ}}^{\text{A}} \cdot \mathbf{q}_{\text{A}}^{\text{B}} \cdot \mathbf{q}_{\text{B}}^{\text{PLM}} \quad (10)$$

according to Eq. (64).

4 The camera reference frame

The right-handed cartesian coordinate system $(X_{\text{CAM}}, Y_{\text{CAM}}, Z_{\text{CAM}})$ is associated with one of the cameras and has its origin exactly in the middle of the 4 CCDs in the associated focal plane.² The Z_{CAM} axis is pointing towards the field-of-view and is called the optical axis, which is usually not parallel with the payload module pointing axis Z_{PLM} as shown in Fig. 3. To define the camera reference frame, we start from the payload module reference frame, and perform the following rotations:

1. we rotate over a position angle η_{CAM} around the Z_{PLM} axis, to obtain the $(X'_{\text{PLM}}, Y'_{\text{PLM}}, Z'_{\text{PLM}})$ reference frame where $Z'_{\text{PLM}} = Z_{\text{PLM}}$. This rotation is illustrated in the left-hand panel of Figure 4, and the corresponding passive rotation matrix is (cf. Eq. (37) and taking the transpose in Eq. (42) already into account):

$$\mathbf{R}_1 = \begin{pmatrix} \cos \eta_{\text{CAM}} & \sin \eta_{\text{CAM}} & 0 \\ -\sin \eta_{\text{CAM}} & \cos \eta_{\text{CAM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (11)$$

2. we rotate over a tilt angle ρ_{CAM} around the Y'_{PLM} axis, to obtain the $(X''_{\text{PLM}}, Y''_{\text{PLM}}, Z''_{\text{PLM}})$ reference frame where $Y''_{\text{PLM}} = Y'_{\text{PLM}}$. This rotation is illustrated in the middle panel of Figure 4 and the corresponding passive rotation matrix is (cf. Eqs. (36) and (42)):

$$\mathbf{R}_2 = \begin{pmatrix} \cos \rho_{\text{CAM}} & 0 & \sin \rho_{\text{CAM}} \\ 0 & 1 & 0 \\ -\sin \rho_{\text{CAM}} & 0 & \cos \rho_{\text{CAM}} \end{pmatrix} \quad (12)$$

3. we rotate over an angle $2\pi - \eta_{\text{CAM}}$ radians around the Z''_{PLM} axis, to obtain the $(X_{\text{CAM}}, Y_{\text{CAM}}, Z_{\text{CAM}})$ reference frame where $Z_{\text{CAM}} = Z''_{\text{PLM}}$. This rotation is illustrated in the right-hand panel of Figure 4 and the corresponding passive rotation matrix is (cf. Eqs. (37) and (42)):

$$\mathbf{R}_3 = \begin{pmatrix} \cos \eta_{\text{CAM}} & -\sin \eta_{\text{CAM}} & 0 \\ \sin \eta_{\text{CAM}} & \cos \eta_{\text{CAM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (13)$$

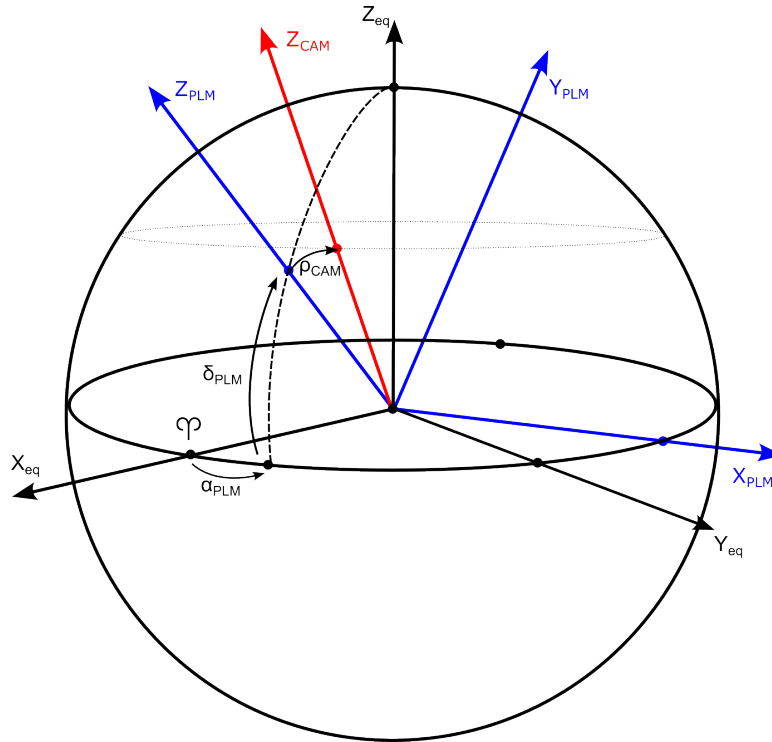


Figure 3: Definition of the tilt angle ρ_{CAM} for the camera reference frame $(X_{CAM}, Y_{CAM}, Z_{CAM})$. Z_{CAM} is the optical axis of the camera, pointing towards the field-of-view.

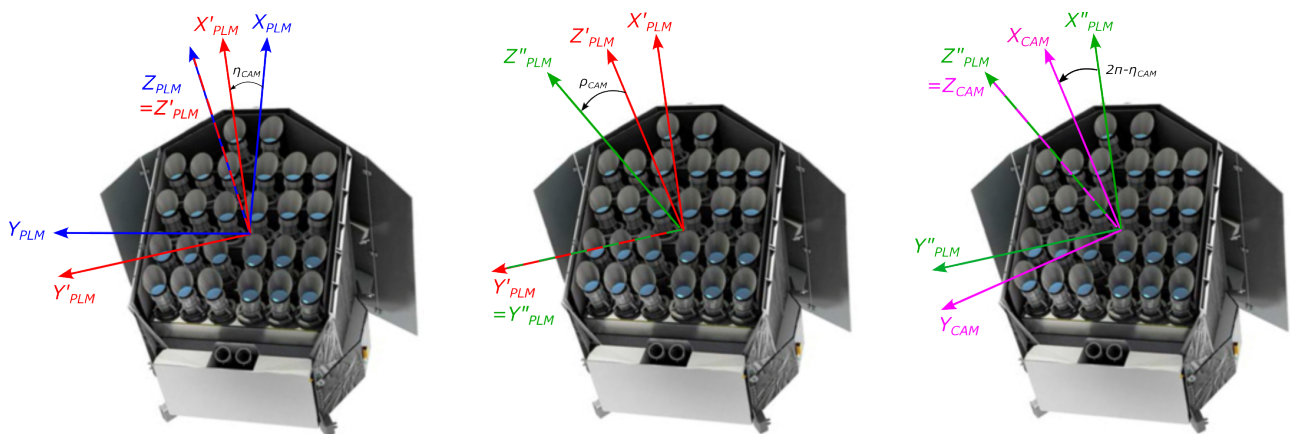


Figure 4: The camera reference frame can be obtained from the payload module reference frame by first rotating the (X_{PLM}, Y_{PLM}) plane around the pointing axis Z_{PLM} over an angle η_{CAM} , then rotating the resulting Z'_{PLM} axis over a tilt angle ρ_{CAM} around the Y'_{PLM} axis, and finally rotating over an angle $2\pi - \eta_{CAM} = -\eta_{CAM}$ around the Z''_{PLM} axis.

A vector with coordinates $(x_{\text{PLM}}, y_{\text{PLM}}, z_{\text{PLM}})$ in the payload module reference frame, has coordinates $(x_{\text{CAM}}, y_{\text{CAM}}, z_{\text{CAM}})$ in the camera reference frame that can be computed using the following transformation:

$$\begin{pmatrix} x_{\text{CAM}} \\ y_{\text{CAM}} \\ z_{\text{CAM}} \end{pmatrix} = \mathbf{R}_{\text{PLM}}^{\text{CAM}}(\eta_{\text{CAM}}, \rho_{\text{CAM}}) \begin{pmatrix} x_{\text{PLM}} \\ y_{\text{PLM}} \\ z_{\text{PLM}} \end{pmatrix} \quad (14)$$

where

$$\mathbf{R}_{\text{PLM}}^{\text{CAM}}(\eta_{\text{CAM}}, \rho_{\text{CAM}}) = \begin{pmatrix} \cos \eta_{\text{CAM}} & -\sin \eta_{\text{CAM}} & 0 \\ \sin \eta_{\text{CAM}} & \cos \eta_{\text{CAM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \varrho_{\text{CAM}} & 0 & -\sin \varrho_{\text{CAM}} \\ 0 & 1 & 0 \\ \sin \varrho_{\text{CAM}} & 0 & \cos \varrho_{\text{CAM}} \end{pmatrix} \begin{pmatrix} \cos \eta_{\text{CAM}} & \sin \eta_{\text{CAM}} & 0 \\ -\sin \eta_{\text{CAM}} & \cos \eta_{\text{CAM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (15)$$

where we used the rotation matrices enumerated above. The inverse transformation is:

$$\begin{pmatrix} x_{\text{PLM}} \\ y_{\text{PLM}} \\ z_{\text{PLM}} \end{pmatrix} = \mathbf{R}_{\text{CAM}}^{\text{PLM}}(\eta_{\text{CAM}}, \rho_{\text{CAM}}) \begin{pmatrix} x_{\text{CAM}} \\ y_{\text{CAM}} \\ z_{\text{CAM}} \end{pmatrix} \quad (16)$$

where

$$\mathbf{R}_{\text{CAM}}^{\text{PLM}}(\eta_{\text{CAM}}, \rho_{\text{CAM}}) = \begin{pmatrix} \cos \eta_{\text{CAM}} & -\sin \eta_{\text{CAM}} & 0 \\ \sin \eta_{\text{CAM}} & \cos \eta_{\text{CAM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \varrho_{\text{CAM}} & 0 & \sin \varrho_{\text{CAM}} \\ 0 & 1 & 0 \\ -\sin \varrho_{\text{CAM}} & 0 & \cos \varrho_{\text{CAM}} \end{pmatrix} \begin{pmatrix} \cos \eta_{\text{CAM}} & \sin \eta_{\text{CAM}} & 0 \\ -\sin \eta_{\text{CAM}} & \cos \eta_{\text{CAM}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (17)$$

is the transpose of the rotation matrix $\mathbf{R}_{\text{PLM}}^{\text{CAM}}(\eta_{\text{CAM}}, \rho_{\text{CAM}})$. The angles η_{CAM} and ρ_{CAM} for the different camera groups of PlatoSim are listed in Table 1. The tilt angle ρ_{CAM} is fixed at 9.2 degrees, and the azimuth angle is $\eta_{\text{CAM}} = 45 + n \cdot 90$ degrees where n is the camera group ID number for the normal cameras.

Camera Group	η_{CAM} (°)	ρ_{CAM} (°)
1	45	9.2
2	135	9.2
3	225	9.2
4	315	9.2
Fast	0	0.0

Table 1: The first column lists the camera group ID number, the second column the azimuth rotation angle η_{CAM} in degrees, and the third column the tilt rotation angle ρ_{CAM} in degrees.

To understand why it is useful to make an extra rotation over the angle $-\eta_{\text{CAM}}$ after rotating first over the angles η_{CAM} and ρ_{CAM} , we first observe that *without* the extra rotation, the location of the spacecraft pointing axis $(x_{\text{PLM}}, y_{\text{PLM}}, z_{\text{PLM}}) = (0, 0, 1)$ in the camera reference frame would be $(x_{\text{CAM}}, y_{\text{CAM}}, z_{\text{CAM}}) = (-\sin \varrho, 0, \cos \varrho)$ (cf. Eq. (14)). That is, in the camera xy-plane the payload module pointing axis would cross the negative x_{CAM} -axis, independent of the camera group. This is illustrated in Fig. 5. The extra rotation allows to align the CCDs of the different camera groups in the sky, and has the additional benefit that the payload module pointing coordinates always fall on (rather than in between) a CCD for each of the 4 camera groups.

²For historical reasons inside PlatoSim's code, the camera reference frame is referred to as the telescope reference frame.

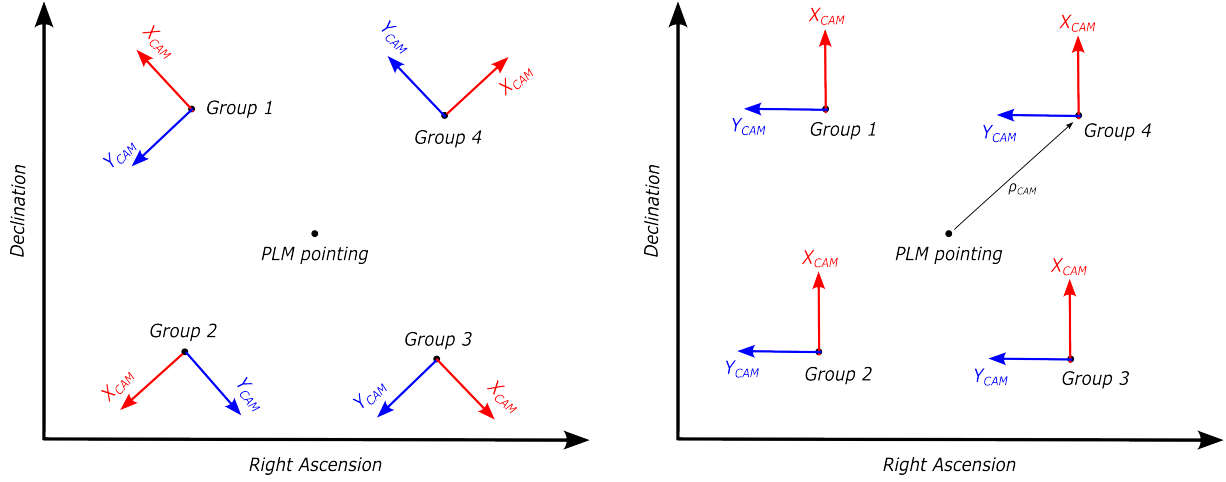


Figure 5: Left: the orientation of the camera reference frame for each of the camera groups when there would be not extra rotation over an angle $-\eta_{CAM}$. Right: the orientation of the same reference frames with the extra rotation as outlined in Eq. (15). ρ_{CAM} is the camera tilt angle. The right-hand side orientation allows for nearly aligned CCDs of the different camera groups on the sky.

5 The Focal Plane reference frame

The right-handed cartesian coordinate system (X_{FP}, Y_{FP}, Z_{FP}) is associated with the Focal Plane of one camera, and like the camera reference frame it has its origin exactly in the middle of the 4 CCDs. The Z_{FP} axis coincides with the Z_{CAM} optical axis of the camera which points towards the field-of-view. The focal plane (X_{FP}, Y_{FP}) is obtained by rotating the (X_{CAM}, Y_{CAM}) plane over an angle γ_{FP} called the focal plane orientation around the $Z_{FP} = Z_{CAM}$ axis. This is illustrated in Fig. 6. The corresponding tranformation (using homogeneous coordinates, see Appendix B.4) is:

$$\begin{pmatrix} x_{FP} \\ y_{FP} \\ f \\ 1 \end{pmatrix} = \mathbf{R}_{CAM}^{FP}(\gamma_{FP}) \begin{pmatrix} x_{CAM} \\ y_{CAM} \\ z_{CAM} \\ 1 \end{pmatrix} \quad (18)$$

where the rotation matrix $\mathbf{R}_{CAM}^{FP}(\gamma_{FP})$ is given by:

$$\mathbf{R}_{CAM}^{FP}(\gamma_{FP}) = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \gamma_{FP} & \sin \gamma_{FP} & 0 & 0 \\ -\sin \gamma_{FP} & \cos \gamma_{FP} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (19)$$

Here $(x_{CAM}, y_{CAM}, z_{CAM}, 1)$ are the homogenized coordinates of a vector in the camera reference frame, and $(x_{FP}, y_{FP}, f, 1)$ are the coordinates of the homogeneous projected vector in the focal plane reference frame. Note that $\mathbf{R}_{CAM}^{FP}$ is not invertible: projecting loses the z_{CAM} information. However, since PlatoSim only uses (sky) directions, the reverse transformation can be obtained by enforcing the length of the vector in the camera reference frame to be unity.

6 The CCD reference frame

The CCD coordinate system (X_{CCD}, Y_{CCD}) is associated with 1 CCD, and has its origin in one of the corners of the CCD. The charge transfer direction during readout of the CCD happens in the negative Y_{CCD} direction towards $y_{CCD} = 0$, and the readout of the serial readout register happens in the negative X_{CCD} direction (towards the origin) for the left side of the CCD, and in the opposite direction for the right side of the CCD. The orientation of the CCD with respect to the (X_{FP}, Y_{FP}) coordinate system is defined by the angle γ_{CCD} , as shown in Fig. 6.

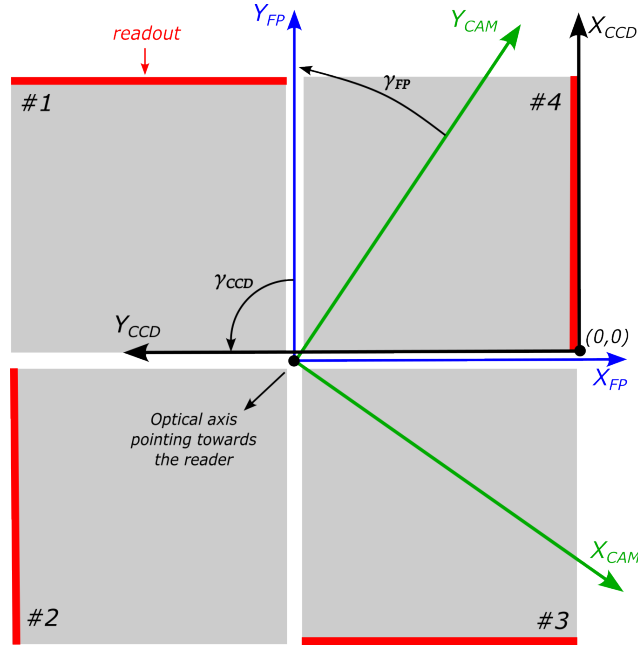


Figure 6: A schematic overview of the focal plane with 4 CCDs. The optical axis Z_{FP} is the blue dot in the middle of the 4 CCDs and points in the positive direction towards the reader. The green axes represent the (x_{CAM}, y_{CAM}) plane in the camera reference frame. γ_{CCD} is the CCD rotation angle.

The transformation (in homogeneous coordinates) between the focal plane and the CCD reference frames is implemented with

$$\begin{pmatrix} x_{CCD} \\ y_{CCD} \\ 1 \end{pmatrix} = \mathbf{R}_{FP}^{CCD}(\gamma_{CCD}) \begin{pmatrix} x_{FP} \\ y_{FP} \\ 1 \end{pmatrix} \quad (20)$$

where

$$\mathbf{R}_{FP}^{CCD}(\gamma_{CCD}) = \begin{pmatrix} \cos \gamma_{CCD} & \sin \gamma_{CCD} & \Delta x_{CCD} \\ -\sin \gamma_{CCD} & \cos \gamma_{CCD} & \Delta y_{CCD} \\ 0 & 0 & 1 \end{pmatrix} \quad (21)$$

where $(\Delta x_{CCD}, \Delta y_{CCD})$ are the coordinates of the CCD zeropoint in the (X_{FP}, Y_{FP}) frame. Note that although we use the same rotation matrix as in Eq. (47), we deviated with the placement of the translation vector for practical reasons. Eq. (20) allows to use the same Δx_{CCD} and Δy_{CCD} for each CCD, independent of the CCD rotation angle γ_{CCD} . An overview of the zero point coordinates and the orientation angle γ_{CCD} is given in Table 2. PlatoSim chooses to define the Focal Plane reference frame such that its axes run parallel with the CCD axes of CCD #3.

CCD	N_{rows}	N_{cols}	r_{exp}	Δx_{CCD} (mm)	Δy_{CCD} (mm)	γ_{CCD} (deg)
1	4510	4510	0	-1.3	82.48	180
2	4510	4510	0	-1.3	82.48	270
3	4510	4510	0	-1.3	82.48	0
4	4510	4510	0	-1.3	82.48	90
1 (F)	4510	4510	2255	-1.3	82.48	180
2 (F)	4510	4510	2255	-1.3	82.48	270
3 (F)	4510	4510	2255	-1.3	82.48	0
4 (F)	4510	4510	2255	-1.3	82.48	90

Table 2: The first column lists the CCD codes: 1,2,3,4 for a normal camera, 1 (F), 2 (F), 3 (F), 4 (F) for a fast camera. r_{exp} is the first row that is exposed; for the fast camera only rows 2255 until 4510 are exposed. Δx_{CCD} and Δy_{CCD} are the CCD origin coordinates in the (X_{FP}, Y_{FP}) frame. γ_{CCD} is the CCD rotation angle.

7 Radial dependency of the PSF

In realistic cases, the PSF is dependent on the angular radial distance θ between the projected center $(x_{\text{FP}}, y_{\text{FP}})$ of the PSF in the focal plane to the optical axis which is assumed to go through the origin $(0, 0)$ of the focal plane. The cosine of this angular distance can be computed using

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{z}}{\|\mathbf{v}\| \|\mathbf{z}\|} \quad (22)$$

where $\mathbf{v} \equiv (x_{\text{FP}}, y_{\text{FP}}, -f)$ is the coordinate vector of the PSF in the pinhole reference frame (see Fig. 9), and $\mathbf{z} = (0, 0, -1)$ is the unit vector along the optical axis but pointing towards the focal plane, also in the pinhole reference frame. This leads to

$$\cos \theta = \left[1 + \left(\frac{x_{\text{FP}}}{f} \right)^2 + \left(\frac{y_{\text{FP}}}{f} \right)^2 \right]^{-1/2} \quad (23)$$

where $(x_{\text{FP}}, y_{\text{FP}})$ are the pinhole-projected coordinates of the center of the PSF, and where f is the focal length of the camera.

8 Payload module jitter rotation angles

We define the payload module yaw φ , the pitch θ , and the roll ψ as rotation angles around respectively the X_{PLM} , Y_{PLM} , and Z_{PLM} axes, such that the angles increase with a clockwise rotation, when looking along the positive axes. This is illustrated in the right-hand panel of Fig. ??.

First a roll rotation is done around the Z_{PLM} axis, then a pitch rotation is done around the rotated Y_{PLM} axis, and finally a yaw rotation is done around the twice-rotated X_{PLM} axis. The combined rotation matrix (in the payload module PLM reference frame) is given by (cf. Eq. (34)):

$$R(\theta, \varphi, \psi) = R(\theta)R(\varphi)R(\psi) \quad (24)$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (25)$$

with the inverse transformation

$$R^{-1}(\theta, \varphi, \psi) = R^t(\psi)R^t(\varphi)R^t(\theta) \quad (26)$$

$$= \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{pmatrix} \quad (27)$$

To compute the updated equatorial sky coordinates $(\tilde{\alpha}_{\text{PLM}}, \tilde{\delta}_{\text{PLM}})$ of the platform Z axis after jitter, we first apply the yaw, jitter, and roll rotations to the current roll axis $\hat{\mathbf{z}}_{\text{PLM}} = (0, 0, 1)$ in the payload module reference frame:

$$\tilde{\mathbf{z}}_{\text{PLM}} = R(\theta, \varphi, \psi) \cdot \hat{\mathbf{z}}_{\text{PLM}} \quad (28)$$

and then use the inverse transformation of Eq. (7) and Eq. 30 to compute the updated $(\tilde{\alpha}_{\text{PLM}}, \tilde{\delta}_{\text{PLM}})$.

Acknowledgements

We thank Denis Griebach of DLR for helpful comments.

A Spherical Coordinate Transformation

We use the following notation for the standard 3D spherical coordinate transformation:

$$f : \mathbb{R}^+ \times [0, \pi] \times [0, 2\pi] \rightarrow \mathbb{R}^3 : \begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \\ z \end{pmatrix} \equiv \begin{pmatrix} r \sin \theta \cos \phi \\ r \sin \theta \sin \phi \\ r \cos \theta \end{pmatrix} \quad (29)$$

and its inverse:

$$f^{-1} : \mathbb{R}^3 \rightarrow \mathbb{R}^+ \times [0, \pi] \times [0, 2\pi] : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \equiv \begin{pmatrix} \sqrt{x^2 + y^2 + z^2} \\ \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \\ \arctan(y/x) \end{pmatrix} \quad (30)$$

B Coordinate transformations

B.1 The components of a rotated vector in the same reference frame

The coordinate transformation in this Section is sometimes called in the literature an *active transformation* or an *alibi transformation*. We consider a vector $\mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y + v_z \mathbf{e}_z$ that is first rotated clockwise by an angle θ around an axis determined by the unit vector $\hat{\mathbf{u}} = u_x \mathbf{e}_x + u_y \mathbf{e}_y + u_z \mathbf{e}_z$, and then translated by a vector $\mathbf{t} = t_x \mathbf{e}_x + t_y \mathbf{e}_y + t_z \mathbf{e}_z$ to give a vector $\mathbf{v}' = v'_x \mathbf{e}_x + v'_y \mathbf{e}_y + v'_z \mathbf{e}_z$. An example is given in Fig. 7 where the rotation axis $\mathbf{u} = \mathbf{e}_z$ is pointing towards the reader. The components of the two vectors are related by:

$$\begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} \quad (31)$$

and the inverse transformation

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R_{\mathbf{u}}^t(\theta) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} - R_{\mathbf{u}}^t(\theta) \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} = R_{\mathbf{u}}^t(\theta) \begin{pmatrix} v'_x - t_x \\ v'_y - t_y \\ v'_z - t_z \end{pmatrix} \quad (32)$$

where the components of the rotation matrix $R_{\mathbf{u}}(\theta)$ are given by the *Rodrigues formula*:

$$R_{ij}(\mathbf{u}, \theta) = \cos \theta \delta_{ij} + (1 - \cos \theta) u_i u_j - \sin \theta \epsilon_{ijk} u_k. \quad (33)$$

where δ_{ij} is the Dirac-delta function and ϵ_{ijk} is the Levi-Civita symbol. Or, written in full:

$$R_{\mathbf{u}}(\theta) = \begin{pmatrix} \cos \theta + u_x^2 (1 - \cos \theta) & u_x u_y (1 - \cos \theta) - u_z \sin \theta & u_x u_z (1 - \cos \theta) + u_y \sin \theta \\ u_y u_x (1 - \cos \theta) + u_z \sin \theta & \cos \theta + u_y^2 (1 - \cos \theta) & u_y u_z (1 - \cos \theta) - u_x \sin \theta \\ u_z u_x (1 - \cos \theta) - u_y \sin \theta & u_z u_y (1 - \cos \theta) + u_x \sin \theta & \cos \theta + u_z^2 (1 - \cos \theta) \end{pmatrix}. \quad (34)$$

$R_{\mathbf{u}}^t(\theta) = R_{\mathbf{u}}^{-1}(\theta)$ denotes the transposed rotation matrix. Looking into the positive direction of $\mathbf{u}$ the rotation angle θ increases clockwise (right-hand rule). It is readily seen that $R_{-\mathbf{u}}(-\theta) = R_{\mathbf{u}}(\theta)$. For the rotation matrices around the principle axes $\mathbf{e}_x$, $\mathbf{e}_y$, and $\mathbf{e}_z$, we obtain

$$R_{\mathbf{e}_x}(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad (35)$$

$$R_{\mathbf{e}_y}(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \quad (36)$$

and

$$R_{\mathbf{e}_z}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (37)$$

Note that the pattern of the minus signs for the case of $R_{\mathbf{e}_y}(\theta)$ differs from that of $R_{\mathbf{e}_x}(\theta)$ and $R_{\mathbf{e}_z}(\theta)$.

Using so-called *homogeneous* coordinates, we can combine the rotation and the translation in a single matrix multiplication:

$$\begin{pmatrix} v'_x \\ v'_y \\ v'_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix}, \quad (38)$$

and the same for the inverse transformation.

$$\begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} v'_x \\ v'_y \\ v'_z \\ 1 \end{pmatrix} = \left(\begin{array}{c|c} R_{\mathbf{u}}^t(\theta) & -R_{\mathbf{u}}^t(\theta) \mathbf{t} \\ \hline 0 & 1 \end{array} \right) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \\ 1 \end{pmatrix}. \quad (39)$$

When applying two successive active rotations using the rotation matrices R_a^b and R_b^c , it is readily seen that $\mathbf{v}_c = R_b^c \cdot (R_a^c \mathbf{v}_a) = (R_b^c \cdot R_a^b) \mathbf{v}_a$. So, for active rotations $a \rightarrow b \rightarrow c$, we have

$$R_a^c = R_b^c R_a^b \quad (\text{active rotation}) \quad (40)$$

where the rotation matrix R_a^b reads as the coordinate transformation when rotating vector a into vector b in the same reference frame.

B.2 The components of a vector in a rotated reference frame

The coordinate transformation in this Section is sometimes called in the literature a *passive transformation* or an *alias transformation*. Given a first reference frame $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, we consider a rotated reference frame of which the principle unit vectors $(\mathbf{e}_{x'}, \mathbf{e}_{y'}, \mathbf{e}_{z'})$ are defined by rotating the principle unit vectors of the first frame, as described in the previous section:

$$(\mathbf{e}_{x'}, \mathbf{e}_{y'}, \mathbf{e}_{z'}) \equiv (R_{\mathbf{u}(\theta)} \mathbf{e}_x, R_{\mathbf{u}(\theta)} \mathbf{e}_y, R_{\mathbf{u}(\theta)} \mathbf{e}_z) \quad (41)$$

Here $R_{\mathbf{u}}(\theta)$ is a rotation matrix given by Exp. (34). The rotation angle θ increases clockwise when looking in the positive direction of the rotation vector $\mathbf{u}$. An example is given in Fig. 8 where the rotation axis $\mathbf{u} = \mathbf{e}_z$ is pointing towards the reader. The components of a vector $\mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y + v_z \mathbf{e}_z = v_{x'} \mathbf{e}_{x'} + v_{y'} \mathbf{e}_{y'} + v_{z'} \mathbf{e}_{z'}$ in the two reference frames are related by

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} = R_{\mathbf{u}}^t(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \quad (42)$$

and

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} \quad (43)$$

In case there is also a translation vector $\mathbf{t} = t_x \mathbf{e}_x + t_y \mathbf{e}_y + t_z \mathbf{e}_z$ involved, we opt to place this vector such that there is consistency with Eqs. (31)-(32).

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} = R_{\mathbf{u}}^t(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} - R_{\mathbf{u}}^t(\theta) \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} = R_{\mathbf{u}}^t(\theta) \begin{pmatrix} v_x - t_x \\ v_y - t_y \\ v_z - t_z \end{pmatrix} \quad (44)$$

and

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \\ t_z \end{pmatrix} \quad (45)$$

Using homogeneous coordinates we can write

$$\begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \\ 1 \end{pmatrix}, \quad (46)$$

and the inverse transformation.

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \\ 1 \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix} = \left(\begin{array}{c|c} R_{\mathbf{u}}^t(\theta) & -R_{\mathbf{u}}^t(\theta) \mathbf{t} \\ \hline 0 & 1 \end{array} \right) \begin{pmatrix} v_x \\ v_y \\ v_z \\ 1 \end{pmatrix}. \quad (47)$$

For successive *passive* rotations using the rotation matrices R_1 and R_2 constructed with Eq. (34), we have $\mathbf{v}' = R_2^t \cdot (R_1^t \mathbf{v}) = (R_1 \cdot R_2)^t \mathbf{v}$ (cf. Eq. (42)) so that for successive rotations we have:

$$R_3 = R_1 \cdot R_2 \quad (\text{passive rotation}) \quad (48)$$

However, potentially confusing, for successive rotations $a \rightarrow b \rightarrow c$, one usually denotes R_1^t with R_a^b and R_2^t with R_b^c so that we still have the usual order:

$$R_a^c = R_b^c \cdot R_a^b \quad (\text{passive rotation}) \quad (49)$$

B.3 Rotations using quaternions

A quaternion $\mathbf{q}$ is a hyper-complex number with four elements, commonly written as

$$\mathbf{q} = q_0 + q_x \mathbf{i} + q_y \mathbf{j} + q_z \mathbf{k} \quad (50)$$

with $(q_0, q_x, q_y, q_z) \in \mathbb{R}^4$. The imaginary numbers $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ follow the following rule:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1 \quad (51)$$

from which follows the so-called right-hand rule, also known as the Hamilton convention:

$$\mathbf{ij} = \mathbf{k}, \quad \mathbf{ji} = -\mathbf{k} \quad (52)$$

This allows two quaternions $\mathbf{q}^a$ and $\mathbf{q}^b$ to be multiplied as

$$\begin{aligned} \mathbf{q}^a \cdot \mathbf{q}^b &= (q_0^a q_0^b - q_x^a q_x^b - q_y^a q_y^b - q_z^a q_z^b) \\ &\quad + (q_0^a q_x^b + q_x^a q_0^b + q_y^a q_z^b - q_z^a q_y^b) \mathbf{i} \\ &\quad + (q_0^a q_y^b - q_x^a q_z^b + q_y^a q_0^b + q_z^a q_x^b) \mathbf{j} \\ &\quad + (q_0^a q_z^b + q_x^a q_y^b - q_y^a q_x^b + q_z^a q_0^b) \mathbf{k} \end{aligned} \quad (53)$$

which is known as the Hamilton product. Note that quaternion multiplication is not commutative. The conjugate $\mathbf{q}^*$ of a quaternion $\mathbf{q} = q_0 + q_x \mathbf{i} + q_y \mathbf{j} + q_z \mathbf{k}$ is defined as $\mathbf{q}^* \equiv q_0 - q_x \mathbf{i} - q_y \mathbf{j} - q_z \mathbf{k}$ and allows to define the norm of a quaternion as

$$\|\mathbf{q}^*\| = \sqrt{\mathbf{q} \cdot \mathbf{q}^*} = \sqrt{q_0^2 + q_x^2 + q_y^2 + q_z^2} \quad (54)$$

Note that $(\mathbf{qp})^* = \mathbf{p}^* \mathbf{q}^*$. The inverse $\mathbf{q}^{-1}$ of a quaternion $\mathbf{q} = q_0 + q_x \mathbf{i} + q_y \mathbf{j} + q_z \mathbf{k}$ can be expressed as

$$\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{\|\mathbf{q}\|^2} \quad (55)$$

so that $\mathbf{q} \cdot \mathbf{q}^{-1} = 1$. For unit quaternions, we have $\mathbf{q}^{-1} = \mathbf{q}^*$.

We consider again a vector $\mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y + v_z \mathbf{e}_z$ that is rotated clockwise by an angle θ around an axis determined by the unit vector $\hat{\mathbf{u}} = u_x \mathbf{e}_x + u_y \mathbf{e}_y + u_z \mathbf{e}_z$, to give a vector $\mathbf{v}' = v'_x \mathbf{e}_x + v'_y \mathbf{e}_y + v'_z \mathbf{e}_z$. This rotation can be characterized by the following unit quaternion:

$$\mathbf{q} = \cos \frac{\theta}{2} + \sin \frac{\theta}{2} (u_x \mathbf{i} + u_y \mathbf{j} + u_z \mathbf{k}) \quad (56)$$

so that

$$\mathbf{v}' = \begin{pmatrix} 0 \\ v'_x \\ v'_y \\ v'_z \end{pmatrix} = \mathbf{q} \mathbf{v} \mathbf{q}^{-1} = \mathbf{q} \mathbf{v} \mathbf{q}^* \equiv \mathbf{q} \begin{pmatrix} 0 \\ v_x \\ v_y \\ v_z \end{pmatrix} \mathbf{q}^* \quad (57)$$

The angle θ increases clockwise when looking in the positive direction of $\mathbf{u}$. Eq. (57) corresponds to the active rotation described in Section B.1. The corresponding rotation matrix $R_{\mathbf{u}}(\theta)$ can be derived from the (unit) quaternion's components as

$$R_{\mathbf{u}}(\theta) = \begin{pmatrix} 2(q_0^2 + q_x^2) - 1 & 2(q_x q_y - q_0 q_z) & 2(q_x q_z + q_0 q_y) \\ 2(q_x q_y + q_0 q_z) & 2(q_0^2 + q_y^2) - 1 & 2(q_y q_z - q_0 q_x) \\ 2(q_x q_z - q_0 q_y) & 2(q_y q_z + q_0 q_x) & 2(q_0^2 + q_z^2) - 1 \end{pmatrix} \quad (58)$$

Given the quaternion $\mathbf{q} = q_0 + q_x \mathbf{i} + q_y \mathbf{j} + q_z \mathbf{k}$, the unit vector $\mathbf{u}$ and the rotation angle θ can be recovered using

$$\mathbf{u} = \frac{q_x \mathbf{e}_x + q_y \mathbf{e}_y + q_z \mathbf{e}_z}{\sqrt{q_x^2 + q_y^2 + q_z^2}} \quad (59)$$

and

$$\theta = 2 \operatorname{atan} \left(\frac{\sqrt{q_x^2 + q_y^2 + q_z^2}}{q_0} \right) \quad (60)$$

In practice the arctangent is computed using the two-argument $\operatorname{atan2}()$ function to identify the correct quadrant of the angle. Note that the quaternion constructed with $\mathbf{u}$ and θ describes the same rotation as the one with $-\mathbf{u}$ and $-\theta$.

Quaternions can also be used to describe passive rotations. As in Section B.2, we are given a first reference frame $(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$, from which we derive a rotated reference frame with principle unit vectors

$$(\mathbf{e}_{x'}, \mathbf{e}_{y'}, \mathbf{e}_{z'}) \equiv (\mathbf{q} \mathbf{e}_x \mathbf{q}^*, \mathbf{q} \mathbf{e}_y \mathbf{q}^*, \mathbf{q} \mathbf{e}_z \mathbf{q}^*) \quad (61)$$

Here $\mathbf{q}$ is quaternion describing the active rotation given by Exp. (56). The rotation angle θ increases clockwise when looking in the positive direction of the rotation vector u . The components of a vector $\mathbf{v} = v_x \mathbf{e}_x + v_y \mathbf{e}_y + v_z \mathbf{e}_z = v_{x'} \mathbf{e}_{x'} + v_{y'} \mathbf{e}_{y'} + v_{z'} \mathbf{e}_{z'}$ in the two reference frames are related by

$$\begin{pmatrix} 0 \\ v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} = \mathbf{q}^{-1} \mathbf{v} \mathbf{q} = \mathbf{q}^* \mathbf{v} \mathbf{q} \equiv \mathbf{q}^* \begin{pmatrix} 0 \\ v_x \\ v_y \\ v_z \end{pmatrix} \mathbf{q} \quad (62)$$

Eq. (62) corresponds to the passive rotation described in Section B.2.

When applying two successive active rotations using the quaternions $\mathbf{q}_1$ and $\mathbf{q}_2$, it is readily seen that $\mathbf{v}' = \mathbf{q}_2(\mathbf{q}_1 \mathbf{v} \mathbf{q}_1^*) \mathbf{q}_2^* = (\mathbf{q}_2 \mathbf{q}_1) \mathbf{v} (\mathbf{q}_2 \mathbf{q}_1)^*$. So, for active rotations $a \rightarrow b \rightarrow c$, we have

$$\mathbf{q}_a^c = \mathbf{q}_b^c \mathbf{q}_a^b \quad (\text{active rotation}) \quad (63)$$

where the quaternion $\mathbf{q}_a^b$ reads as the coordinate transformation when rotating vector a into vector b in the same reference frame. For successive *passive* rotations using the quaternions $\mathbf{q}_1$ and $\mathbf{q}_2$, we have $\mathbf{v}' = \mathbf{q}_2^*(\mathbf{q}_1^* \mathbf{v} \mathbf{q}_1) \mathbf{q}_2 = (\mathbf{q}_1 \mathbf{q}_2)^* \mathbf{v} (\mathbf{q}_1 \mathbf{q}_2)$, implying that for passive rotations $a \rightarrow b \rightarrow c$, we have

$$\mathbf{q}_a^c = \mathbf{q}_a^b \mathbf{q}_b^c \quad (\text{passive rotation}) \quad (64)$$

In practice care has to be taken, however, as sometimes $\mathbf{q}_a^b$ denotes $\mathbf{q}_1^*$ rather than $\mathbf{q}_1$ which changes the order of the quaternions.

B.4 Pinhole projection

The standard pinhole camera model is shown in Fig. 9. The Z axis is the optical axis of the camera, the (X, Y) plane is parallel to the focal plane, but has its origin in the pinhole. The focal plane (with the CCDs) is at a distance f behind the pinhole, where f is the focal length of the camera. Light rays of a star at point P go through the pin hole O and are projected to a point Q on the focal plane. In the (Y, Z) plane, the triangles $\overline{OP'P''}$ and $\overline{OQ'Q''}$ are similar triangles, so that we have $|P'P''|/|OP'| = |Q'Q''|/|OQ'|$ or

$$\frac{-y_q}{f} = \frac{y_p}{z_p} \quad (65)$$

where the minus sign indicates that the pinhole reverses the image on the focal plane. A similar argument holds in the (X, Z) plane, so that we can write

$$\begin{pmatrix} x_q \\ y_q \\ -f \end{pmatrix} = -\frac{f}{z_p} \begin{pmatrix} x_p \\ y_p \\ z_p \end{pmatrix} \quad (66)$$

First note that the units of (x_q, y_q) are independent of the units of (x_p, y_p, z_p) and are completely determined by the units of the focal length f . The latter could be expressed in milli-meter, pixels, or one can use normalized coordinates by setting $f = 1$. Secondly, the same projected point (x_q, y_q) is obtained by any point $(\alpha x_p, \alpha y_p, \alpha z_p)$ with $\alpha \in \mathbb{R}_0$.

With normal coordinates, the non-linear division by z_p prevents expressing the pinhole projection as a convenient matrix multiplication. We therefore go to augmented space and use homogeneous coordinates $(x_q, y_q, 1)$ and $(x_p, y_p, z_p, 1)$

$$\begin{pmatrix} x_q \\ y_q \\ z_q \\ 1 \end{pmatrix} \sim \mathbf{C} \begin{pmatrix} x_p \\ y_p \\ z_p \\ 1 \end{pmatrix} \quad (67)$$

where the equality $=$ has been replaced by the equivalence relation $\sim$. After any transformation a homogenisation of the coordinates is executed, which means dividing all coordinates with the fourth one. $\mathbf{C}$ is called the *camera matrix*, for which different versions circulate in the literature. For example:

$$\mathbf{C}_1 = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (68)$$

is the augmented equivalent of Eq. (66). The following version avoids mirroring the x - and y - axes of the projected reference frame:

$$\mathbf{C}_2 = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (69)$$

Neither $\mathbf{C}_1$ and $\mathbf{C}_2$ are invertible, as the projection on a plane loses the original z_p coordinate. Alternatively, the following camera matrix retains this information, and is invertible:

$$\mathbf{C}_3 = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & f & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (70)$$

the resulting homogenised coordinate $z_q = 1 + 1/z_q$ coordinate is never used in practice, but simply guarantees invertibility of the matrix.

